

AUDUBON COUNTY AP, IA, USA

WMO: 725498

Lat: 41.700N

Lon: 94.917W

Elev: 392

StdP: 96.70

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94998

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-17.8	-25.2	0.4	-19.7	-22.5	0.5	-17.3	11.5	-3.9	10.4	-3.6	1.6	320	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	33.0	24.6	32.2	24.1	30.8	23.6	26.7	31.2	25.5	29.8	24.5	28.9	4.6	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	21.4	29.8	24.0	19.8	28.5	22.8	18.4	27.6	86.6	31.0	80.4	30.0	76.1	28.9	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	8.7	7.9	-26.0	36.1	2.8	1.9	-28.0	37.5	-29.7	38.6	-31.3	39.6	-33.3	41.0		
			-26.3	28.2	2.6	1.1	-28.1	29.0	-29.6	29.7	-31.1	30.3	-33.0	31.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	-5.5	-4.4	3.1	9.7	15.9	21.4	23.4	21.6	18.0	10.5	3.2	-3.4	(6)
	DBStd	11.54	6.58	6.70	6.66	5.47	4.81	3.56	3.19	3.18	4.68	5.26	5.81	6.26	(7)
	HDD10.0	1883	481	403	231	71	7	0	0	0	2	58	214	416	(8)
	HDD18.3	3677	739	637	472	263	106	12	2	8	63	248	453	674	(9)
	CDD10.0	1713	0	0	18	61	191	342	414	361	241	74	11	1	(10)
	CDD18.3	464	0	0	1	4	31	104	159	110	52	5	0	0	(11)
	CDH23.3	4811	0	0	7	72	332	1064	1659	1051	557	66	2	0	(12)
	CDH26.7	1686	0	0	1	16	98	374	649	353	182	13	0	0	(13)
(14) Wind	WSAvg	3.1	3.4	3.5	3.8	4.0	3.4	2.9	2.2	2.0	2.5	3.0	3.4	3.2	(14)
(15) Precipitation	PrecAvg	874	22	27	55	93	135	121	118	101	72	62	43	25	(15)
	PrecMax	1153	49	63	134	194	301	311	260	225	201	201	109	71	(16)
	PrecMin	654	7	3	3	19	33	20	35	26	19	12	2	0	(17)
	PrecStd	156	11	18	38	47	67	74	61	53	44	44	30	18	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.5	14.0	23.9	29.1	32.7	34.0	35.3	35.0	32.9	28.0	22.5	14.0	(19)
		MCWB	6.1	8.5	15.2	18.1	20.8	22.9	26.5	24.7	23.3	18.0	14.8	10.1	(20)
	2%	DB	7.9	10.3	20.9	25.9	29.1	32.4	33.0	32.3	31.1	25.1	18.0	10.2	(21)
		MCWB	4.5	7.0	13.8	16.0	20.9	23.3	25.7	24.3	23.0	16.4	13.0	7.2	(22)
	5%	DB	5.3	7.4	17.2	22.6	27.3	31.0	32.3	30.2	28.7	22.4	15.2	7.6	(23)
		MCWB	2.5	4.6	12.0	14.3	18.9	22.6	25.3	24.3	21.9	16.0	11.0	5.0	(24)
	10%	DB	2.9	4.1	13.7	19.9	24.9	28.8	30.2	28.0	27.1	19.9	12.6	5.1	(25)
		MCWB	1.2	2.2	9.5	13.0	17.6	21.6	24.5	22.7	20.4	14.6	8.6	2.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	9.8	17.5	20.0	24.1	26.8	28.6	27.5	25.6	21.2	16.3	12.1	(27)
		MCDB	10.8	12.4	21.5	25.7	28.8	31.3	32.9	31.1	30.6	23.8	19.9	13.5	(28)
	2%	WB	4.9	7.0	15.0	17.6	22.2	25.1	27.4	25.8	24.1	18.9	14.0	7.7	(29)
		MCDB	8.3	10.1	18.1	22.9	27.7	29.7	31.9	30.3	28.7	23.1	17.3	9.2	(30)
	5%	WB	3.1	4.4	12.6	15.7	20.3	24.0	26.1	24.8	23.0	16.9	11.5	5.0	(31)
		MCDB	5.3	7.2	16.4	21.0	25.3	28.8	30.3	29.1	27.5	20.8	14.8	7.7	(32)
	10%	WB	1.3	2.5	9.5	13.9	18.9	22.9	24.9	23.7	21.5	15.0	8.9	3.0	(33)
		MCDB	3.1	4.5	13.9	18.8	23.5	27.5	29.2	27.8	25.4	19.0	12.4	5.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.1	11.3	12.9	12.2	11.9	12.1	12.1	13.7	12.7	11.8	9.5	(35)
		MCDBR	13.0	13.1	16.6	17.9	14.9	13.3	12.8	13.9	15.1	16.4	16.2	13.1	(36)
	5% WB	MCWBR	10.1	9.9	10.3	10.1	7.8	6.8	7.2	8.2	8.5	9.6	10.8	10.3	(37)
		MCWBR	12.0	12.3	14.1	14.3	12.7	11.4	11.4	12.0	13.0	13.2	14.5	11.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.284	0.310	0.323	0.378	0.402	0.413	0.429	0.415	0.380	0.334	0.294	0.284	(40)
		taud	2.416	2.354	2.401	2.269	2.273	2.296	2.238	2.286	2.340	2.456	2.497	2.437	(41)
	Ebn at Noon		872	896	926	890	874	862	844	845	852	859	855	839	(42)
		Edh at Noon	78	98	105	129	132	130	136	126	111	88	72	71	(43)
(44) All-Sky Solar Radiation	RadAvg		1.95	2.95	3.80	4.79	5.70	6.32	6.41	5.52	4.58	3.13	2.15	1.62	(44)
	RadStd		0.18	0.19	0.36	0.37	0.43	0.54	0.44	0.38	0.31	0.38	0.16	0.14	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page